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Private Property Rights and Sharecropping

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The purpose of this study is to derive a theory of share tenancy with which to explore the nature of resource allocation under one of the main forms of land tenure in agriculture. Share tenancy is a land lease under which rent is a contracted percentage of the output yield from the tenant per period of time. As a rule, the landowner provides land, and the tenant provides labor; other inputs may be provided by either party. Share tenancy (or sharecropping) is thus share contracting, defined here as two or more individual parties combining privately owned resources for the production of certain mutually agreed outputs, the actual outputs to be shared according to certain mutually accepted percentages as returns to the contracting parties for their productive resources forsaken. The theory, to be derived from standard economic principles, may be generalized to all forms of land tenure under similar ownership of resources.

The prevailing impression is that sharecropping results in inefficient allocation of resources.¹ It will be shown here that the inefficiency argument is illusory. The implied resource allocation under private property rights is the same whether the landowner cultivates the land himself, hires farmhands to do the tilling, leases his holdings on a fixed rent basis,

* When this paper was written, the author was assistant professor at California State College, Long Beach. The paper is primarily drawn from Chapter ii of a lengthier manuscript to be published by the University of Chicago Press (Cheung, in press). Thanks for financial support are given to the Lilly Endowment Incorporated grant to UCLA for the study of property rights and behavior.

¹ Classical economists aside, contemporary theses on share tenancy which imply inefficient resource allocation include Schickele (1941), Heady (1947), Heady and Kehrberg (1952), Issawi (1957), Georgescu-Roegen (1960), and Sen (1966). On the other hand, Johnson (1950) is reluctant to accept the inefficiency argument—in spite of his theoretical model which supports it. Bray (1963) argues that, theory aside, agricultural yields under share leases have been impressive in the United States. Condemnation of sharecropping on efficiency grounds can be traced back to Adam Smith, if not earlier. It is interesting to note that classical writers were more diversified in views and on occasions came closer to the model offered in this study than modern writers (see Cheung, in press, chap. iii).

or shares the actual yield with his tenant. Before presenting my theoretical solution, I shall discuss briefly the prevailing analysis of sharecropping.

I. The Tax-equivalent Approach Examined

The traditional approach to analyzing share tenancy may be called the "tax-equivalent approach." Since under sharecropping a portion of every output unit produced is taken away as rent, this yields the impression of being similar to an *ad valorem* excise tax—where part of every unit produced is "taxed" away by the landowner (government). The distribution of output is not the same, it is believed, as with fixed rent or owner cultivation—where the tiller obtains the *entire* incremental product. Share tenancy, therefore, is said to result in less intensive (and less efficient) farming because the tenant's incentive to work or invest in land is reduced.

To clarify this line of argument, a geometric presentation will suffice.² For simplicity, let us assume that the only tenant input is labor. In Figure 1,

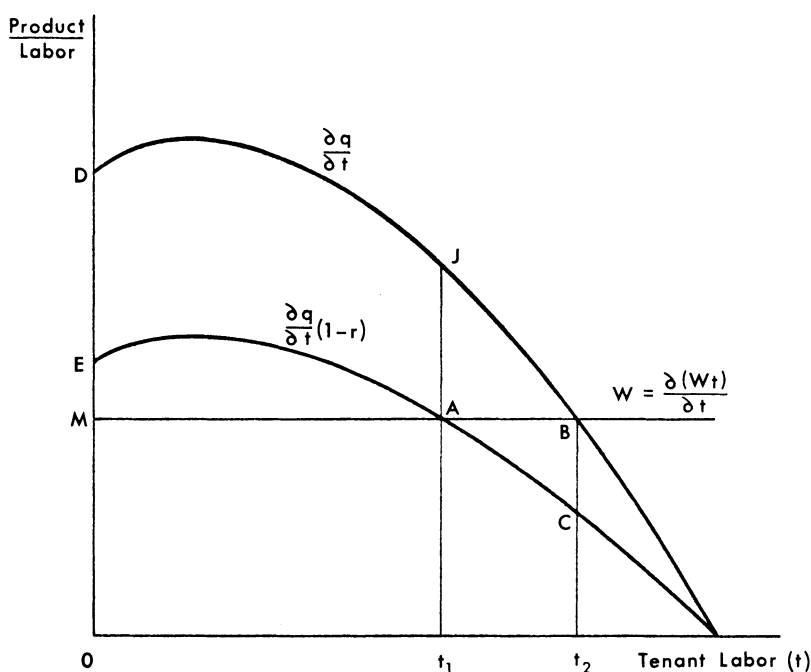


FIG. 1.—The tax-equivalent approach illustrated

² The analogy of share rent to an excise tax is explicit in Smith (1937, p. 367). The geometry in Fig. 1, which employs marginal analysis, is derived by tracing through footnotes in two chapters of Marshall (1956, Book IV, chap. iii; and Book VI, chap. x). As subsequent writers generally adopted Marshall's analysis of sharecropping they appear to have overlooked footnote 2 in *ibid.*, p. 536, where Marshall considered the possibility of efficient sharecropping.

tenant labor, t , is measured along the horizontal axis, and $\partial q/\partial t$ represents the marginal product of tenant labor *with a given plot of land*. The marginal tenant cost, $\partial(Wt)/\partial t$, is horizontal under a competitive market, with W as the prevailing wage rate. If the landowner is to hire farmhands to do the tilling, equilibrium is at B , and the quantity of farm labor hired will be t_2 . At this equilibrium, we have the marginal equality: $\partial q/\partial t = \partial(Wt)/\partial t$. The same result obtains for owner cultivation, whether the owner works up to t_2 on his holding and works elsewhere, or whether he works to less than t_2 and hires additional laborers. The total rent as a return to land received by the landowner is represented by the area MDB , an amount equal to that of a fixed-rent contract.

Under the tax-equivalent approach of analyzing share tenancy, however, the marginal tenant receipt net of rent, $(\partial q/\partial t)(1 - r)$, will shift *downward* at every point. That is, if the landowner takes 40 per cent of the annual yield ($r = 0.4$) and the tenant takes 60 per cent, $(\partial q/\partial t)(1 - r)$ will be 60 per cent of $\partial q/\partial t$ at every point. With the tenant's decision made at the margin, it is said, equilibrium is at A , where the marginal tenant cost equals the marginal tenant receipt; that is, $\partial(Wt)/\partial t = (\partial q/\partial t)(1 - r)$. The associated quantity of tenant labor will be t_1 . Under this condition, the total product is represented by the area $ODJt_1$, with the landowner getting a rent equaling area $EDJA$ and the tenant's share equaling area $OEAt_1$. With equilibrium at A , the marginal product of tenant labor is higher than the marginal tenant cost. Share tenancy is therefore inefficient (with area JAB representing the economic waste). Similar conclusions have been reached for other non-land inputs and investment in land.

It is not difficult to see that the so-called equilibrium at position A is only asserted. As shown in Figure 1, area MEA represents the amount received by the tenant over and above his alternative earning (area $OMAt_1$). The existence of this residual earning for the tenant (area MEA) is inconsistent with equilibrium. Under private ownership of land, the landowner can extract the residual earning from the tenant or he can stipulate in the share contract an amount of tenant input larger than t_1 . Furthermore, the landowner could have chosen instead a wage contract, a fixed-rent contract, or to sell his landownership outright, and in each case the rental (interest) income would be higher than share rent. Indeed, even if we accept the tax-equivalent approach and let the tenant *alone* make the decision at the margin, there is no reason for him to choose to work up to t_1 for one landowner. The tenant's total income from farming would be higher should he work in many farms and commit a trivial amount of labor in each farm.³

³ Under private ownership of resources, it is difficult, if not impossible, to define a set of constraints which makes position A in Fig. 1 an equilibrium. Under state ownership of land, however, an interesting set of conditions for position A can be specified: (a) that land ownership belongs to the state, and the government arbitrarily

The looseness of the tax-equivalent analysis stems from the lack of any explicit treatment of the terms in a share contract which the participating parties must *mutually* agree to abide by when the contract is formed. Under a fixed-rent or a wage contract, the contractual terms of factor payments are variables determined in the market, and they in turn determine the distribution of income for the contracting parties and resource allocation. The share contract is no exception, although its contractual terms are expressed in different dimensions. In such light, I offer below an analysis which yields the terms of share contracting that are consistent with market equilibrium.

II. The Solution Defined

Our analysis is based on the premise of wealth maximization subject to the constraints of private property rights in a free market. Under exclusive and transferable resource rights, each contracting party is free to accept or reject the negotiated terms of a share contract. We assume a zero cost of contracting.⁴

In Figure 2, the abscissa of the vertical supply curve, S , indicates the total area of land *belonging to a landlord*. Let h denote the land area held by a tenant, and q be the product. The marginal product of land, $\partial q/\partial h$, diminishes as h increases, holding the farming inputs of one tenant (or one tenant family) constant. Suppose the rent charged by the landowner is 60 per cent of the annual yield, that is, $r = 0.6$. A marginal contract rent curve, $(\partial q/\partial h)r$, is at 60 per cent of $\partial q/\partial h$. The vertical distance between $\partial q/\partial h$ and $(\partial q/\partial h)r$ is the marginal tenant income, $(\partial q/\partial h)(1 - r)$, defined as the change in tenant income with respect to a change in land area used by the tenant. The shaded area between $\partial q/\partial h$ and $(\partial q/\partial h)r$ represents the total farming income received by the tenant, and the area below $(\partial q/\partial h)r$ represents the total rent collected by the landowner. If the tenant's income is higher than his alternative earning, he will stay in farming and will use all the land available as long as the marginal productivity of land

assigns land to tenants with rents collected on a share basis; (b) that the tenant is free to work elsewhere at a market wage rate, and the amount of labor he uses on the state land is not stipulated; and (c) that the tenant's lease right over the state land is not transferable in any form. If strictly enforced, it appears that equilibrium will be at A , and the residual earnings for the government tenants are similar to "welfare" payments. Yet this is hardly the set of constraints which writers on sharecropping have in mind.

⁴ Insofar as there is more than one individual wanting the same property resource, *competition* is implied, and the number of competitors includes not only those who are actually using the resource but also potential owners (users). The assumption of zero contracting cost substitutes for the sometimes dubious assumption of "pure" competition. Included in the general term "contracting cost" are the costs of negotiating and the costs of enforcing the stipulations of the contract. For a discussion of contracting with non-zero transaction costs, see Cheung (in press, chap. iv).

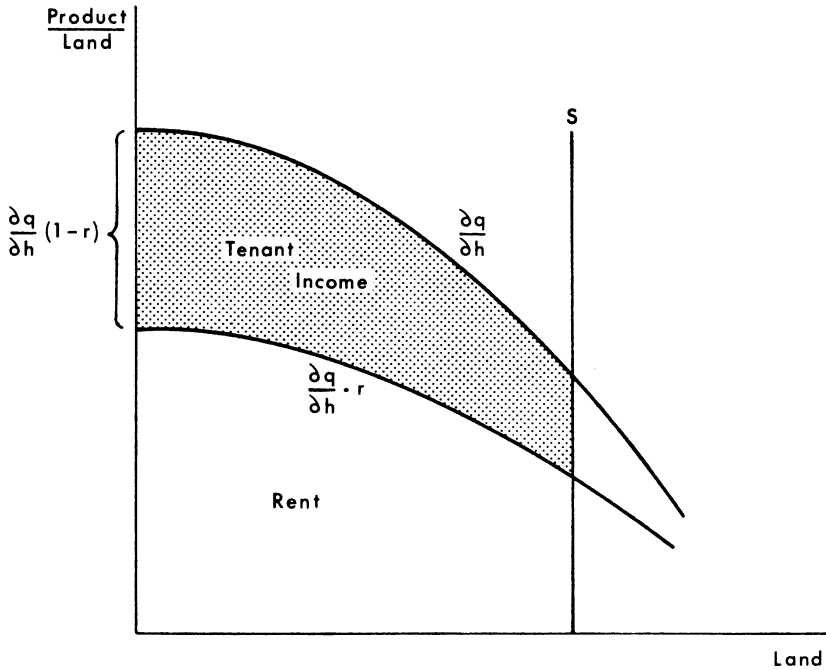


FIG. 2.—Sharecropping with one tenant

is greater than zero, with all farming inputs other than land held constant. To maximize wealth, the landowner will raise the rental percentage—thus the $(\partial q/\partial h)r$ curve—until the tenant's income from farming equals his alternative earning.

However, the rental percentage is not the only variable which the landowner can adjust to maximize his wealth. The landlord will not allow one tenant to cultivate all the land he owns if parceling his land to several tenants will result in a higher total rent. This is illustrated in Figure 3. In this figure, vertical lines $T_1, T_2, T_3, \dots$ are dividing lines of plots of land to the first, second, and third tenant, respectively. As the number of tenants cultivating the available land increases, the marginal product of land shifts upward relative to the situation where there is only one tenant. Assume for the moment that the rental percentage is the same for all tenants. The curves $(\partial q/\partial h)_1, (\partial q/\partial h)_2, \dots$ are the marginal productivities of land for each tenant, and $(\partial q/\partial h)_1 r, (\partial q/\partial h)_2 r, \dots$ are the respective marginal contract rent curves for each tenant.⁵ The income for each tenant is represented by the area between the respective $\partial q/\partial h$ and $(\partial q/\partial h)r$ curves

⁵ Depending on the production function, these pairs of curves may not be identical. We shall discuss this later.

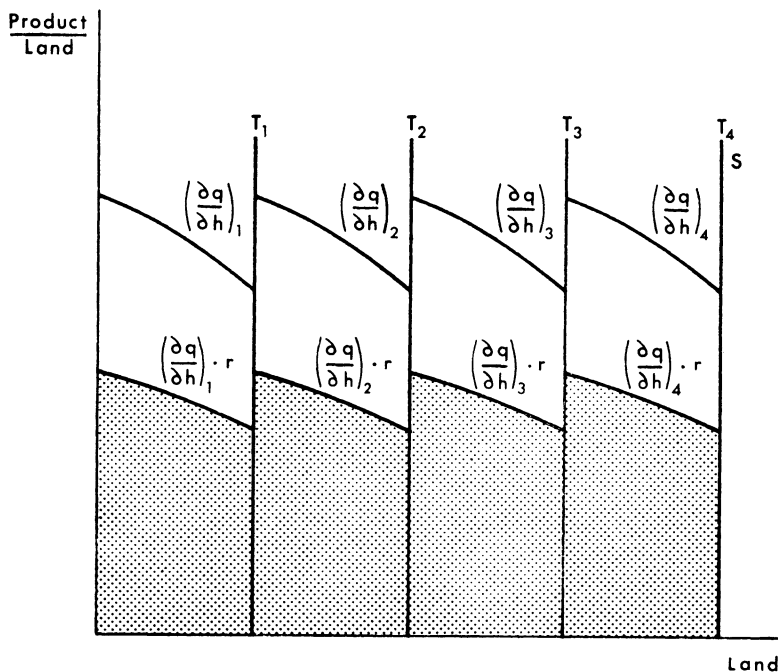


FIG. 3.—Sharecropping with multiple tenants

for the tenant. To maximize his wealth, the landowner will maximize the *difference* between the integral of the marginal productivities of land and the integral of the tenants' incomes, which is represented by the shaded area in Figure 3. That is, the landowner will make the integral of marginal contract rents a maximum. This implies that the income of each tenant will not be higher than his alternative earning.

As the area of land assigned to each tenant becomes smaller, however, the rental percentage that the landlord charges must be lower for the tenant to obtain his alternative earning, which calls for a decrease in marginal contract rent, $(\partial q/\partial h)r$, in order to prevent the tenant from giving up the lease. *This decrease in $(\partial q/\partial h)r$ will lead to a lower rent received from each tenant, and if the land size per tenant continues to decrease, the rental percentage will eventually become so low that the total rent from the land will decline.* The solution is thus well defined: Maximization of the landowner's wealth, given his total landholdings and the cost of tenant inputs, requires the simultaneous determination of the land size per tenant and the rental percentage. In other words, with private property rights over land and tenant input resources, the terms in a share contract mutually agreed upon by the landowner and the tenant will include the rental percentage and the ratio of non-land to land input that are consistent with equilibrium.

III. The Solution in Mathematics

For simplicity of presentation, assume there are two *homogeneous* factors of production: the amount of land per tenant farm (h) and the amount of tenant labor per farm (t). Further assume that the production functions of each tenant farm are identical. With these assumptions, it follows that the land size, h , and the rental percentage, r , contracted for each tenant farm will be the same in equilibrium.

Let each tenant farm's production function be $q = q(h, t)$. The amount of land per farm, h , is equal to the total landholding of the landowner, H , divided by the number of farms, m ; that is, $h = H/m$. The landowner's total rent, R , is then equal to the number of farms times the rent per farm, that is, $R = m \cdot r \cdot q(h, t)$. Under competition, $Wt = (1 - r)q(h, t)$, where W is the market wage rate of the tenant labor, t .

The problem of the landowner is then to maximize R , through the choice of m , r , and t , subject to the constraint of competition;⁶ that is,

$$\max_{(m, r, t)} R = m \cdot r \cdot q(h, t), \quad \text{subject to} \quad Wt = (1 - r)q(h, t).$$

Forming the Lagrangian expression, the problem is thus the maximization of $L = m \cdot r \cdot q(h, t) - \lambda[Wt - (1 - r)q(h, t)]$. The necessary conditions are:

$$\frac{\partial L}{\partial m} = r \cdot q(h, t) + m \cdot r \cdot \frac{\partial q}{\partial h} \frac{dh}{dm} + \lambda(1 - r) \frac{\partial q}{\partial h} \frac{dh}{dm} = 0. \quad (1)$$

$$\frac{\partial L}{\partial r} = m \cdot q(h, t) - \lambda q(h, t) = 0. \quad (2)$$

$$\frac{\partial L}{\partial t} = m \cdot r \cdot \frac{\partial q}{\partial t} - \lambda W + \lambda(1 - r) \frac{\partial q}{\partial t} = 0. \quad (3)$$

$$\frac{\partial L}{\partial \lambda} = -Wt + (1 - r)q(h, t) = 0. \quad (4)$$

From equation (2) above, we have

$$\lambda = m.$$

And noting that $dh/dm = d(H/m)/dm = -H/m^2$, equation (1) becomes

$$r \cdot q + m \cdot r \cdot \frac{\partial q}{\partial h} \left(\frac{-H}{m^2} \right) + m(1 - r) \frac{\partial q}{\partial h} \left(\frac{-H}{m^2} \right) = 0;$$

that is,

$$r \cdot q - \frac{h \partial q}{\partial h} = 0; \quad \text{or} \quad \frac{rq}{h} = \frac{\partial q}{\partial h}.$$

This indicates that *the rent per acre of land equals the marginal product of land in equilibrium*, a condition identical to that of a fixed-rent contract.

⁶ Note that t and m need not be treated separately. Given t , an adjustment of m yields the same result as adjusting t by holding m constant. They are separated here for the purpose of conveniently deriving all of the wanted conditions in equilibrium.

space owned by the landlord may not be exhausted. The curve q/h represents the average product of land with one tenant family being employed, that is, the average product with respect to the land size while holding all other farming inputs (of one tenant family) constant. The curve f/h , or the *fixed total* farming cost divided by land area, reveals the cost of farming inputs (other than land) which yield the expected q/h . Assume for the moment that all farming inputs are borne by the tenant; the f/h curve is the total cost other than land divided by the respective land space. It includes the costs of labor, seeds, fertilizers, farming equipment, and so forth, associated with the period of time of the relevant production run.⁷ That is, $f/h = (p_t t + p_z z + \dots)/h$; where f is the *constant* total cost other than land, and p_t , p_z , $\dots$ are the factor prices of tenant labor, t ; fertilizers, z ; $\dots$. Since we hold the non-land farming inputs constant, the f/h curve is a rectangular hyperbola. The vertical difference between q/h and f/h defines $(q - f)/h$, the *rent per unit of land*, taking into account the alternative cost of the tenant.⁸

The total amount of tenant inputs which define f/h are contractually stipulated, which is essential because the tenant would commit less if only the rental percentage were prescribed. As shown in Section I, given any rental percentage, for every unit of output produced, only a portion of it will go to the tenant. If farming decisions *were made entirely by the tenant*, it would be to his interest that the cost of incremental tenant input be less than the associated marginal product.

Under mutual agreements, economic theory implies that the total amount of tenant inputs contracted will be the amount which yields the highest $(q - f)/h$, or which yields the highest rent per unit of land. Since in Figure 4 the *marginal* farming cost always equals zero, the highest value of $(q - f)/h$ can be derived as follows. For every upward *shift* of f/h as a result of increasing the stipulated amount of tenant input, there is associated an upward *shift* of q/h . The former represents the marginal non-land cost, which increases at a constant rate (that is, constant marginal factor cost under pure competition); the latter represents the marginal productivity of additional tenant (that is, non-land) input, which increases at a decreasing rate (that is, decreasing marginal product of tenant input).⁹ The highest $(q - f)/h$ —associated with a specific f/h curve—is obtained

⁷ The important distinction here is between land and inputs other than land. The substitution between labor and farming equipment, for example, is not yet important in this theory and is ignored.

⁸ For simplicity of presentation, all values are measured in real terms. Note, however, that the annual yield, q , may include a variety of crops; given their relative market prices, the value of each may be expressed in terms of one crop, say, rice. And payments for tenant inputs and land will likewise be in rice.

⁹ To avoid semantic problems, we assume here the increase of *one* tenant input only. If several inputs are increased, as is usually the case, the relevant “shifts” to consider are when q/h increases at a decreasing rate. In either case, the theoretical outcomes are identical.

when the marginal upward shifts of f/h and q/h are equal, or when the marginal product of tenant input equals the marginal tenant cost. The associated non-land farming cost, which defines f/h , covers the level of tenant inputs consistent with productive equilibrium. To maximize wealth, the crops chosen to be planted or rotated, and the method of producing them, are, by definition, those which yield the highest present value of land to the landowner, that is, those which maximize *rental annuity*.¹⁰ The relevant $(q - f)/h$ curve, or average rent, for decision making is thus the highest one derived from alternative pairs of q/h and f/h . To state it more precisely, the highest value of $(q - f)/h$ defines the cost of land per acre as a factor of production.¹¹

The marginal product of land, $\partial q/\partial h$ —or $(\partial q/\partial h)_1$ in Figure 3—cuts q/h and $(q - f)/h$ at their highest points. The equilibrium land size assigned to this tenant, T_1 , is where $(q - f)/h$ is at a maximum. Maximizing rent per acre of land will maximize the rental annuity from the landlord's total landholdings. With the equilibrium tenant land size determined (T_1), the equilibrium rental percentage equals $(q - f)/h$ divided by q/h (at T_1). That is, the rental percentage, r , equals ar/ap as labeled in Figure 4. Given this equilibrium percentage, say 70 per cent, we plot the marginal contract rent curve, $(\partial q/\partial h)r$, as this percentage of $\partial q/\partial h$ at every point.

Since the tenant contracts to pay a percentage of the total product, as indicated by $(\partial q/\partial h)r$, the cost of land is no longer a constraint with respect to the amount of land the tenant will use. To maximize his income, the tenant prefers to employ land to the point where $\partial q/\partial h$ is zero (while holding his farming inputs constant as contractually stipulated). The landowner, on the other hand, will limit the tenant's holding to T_1 and parcel the remainder of his total holding to other tenants under similar contractual arrangements.¹² The landowner cannot successfully restrict

¹⁰ Given the appropriate discount rate, ρ , and the future income stream from the land, $y_1, y_2, \dots, y_n$, with y_i not necessarily equal to y_j , there exists a $\bar{y}$, called the *rental annuity*, such that

$$V_0 = \bar{y} \left[\frac{(1 + \rho)^n - 1}{(1 + \rho)^n} \right] = \frac{y_1}{1 + \rho} + \frac{y_2}{(1 + \rho)^2} + \dots + \frac{y_n}{(1 + \rho)^n}.$$

If the private land right is perpetual, we have $V_0 = \bar{y}/\rho$, where V_0 is the present value of land.

¹¹ The notion that rent is the cost of land began perhaps with Marshall. And I maintain here that it is the *highest* rent per unit of land which really counts, for it is from this that the land value is determined.

¹² It remains to show how the last plot of land belonging to the same landlord is allocated, since it may be too small for one tenant family. One answer is that a tenant may be renting lands from two or more landlords at the same time, employing land margins from both. Likewise, a tenant may own a small area of land himself, renting an extra margin from a landlord. This is evident in the existence of part-owner farmers. Therefore, if the land size of the last plot assigned to a tenant family is smaller than the equilibrium size, the tenant will employ part of his resources elsewhere, possibly outside the farming sector.

land size to less than T_1 , since with rental percentage r the tenant's alternative earning will be higher elsewhere, and he will give up the tenancy.

Several side remarks can be made. First, not all tenant families are equally productive. Some will be able to produce more because of the endowments of certain "specific" factors of production, for example, varying possession of knowledge. In competitive equilibrium, the more productive tenants will be the intramarginal tenants, and the f/h curve thus defined will include the imputed rent for the tenant. The landowners cannot successfully discriminate among tenants of varying efficiencies even if there were no costs associated with discrimination, since landlords employing marginal tenants (least productive tenants) would bid the more productive tenants away from a discriminating landlord.

Second, even if land is homogeneous, the land size per tenant may not be the same. Defining land homogeneity as physically identical and having the same rental value per unit of land—that is, the $(q - f)/h$ curves in different tenant farms yield equal heights at their peaks—the land sizes for different tenants will differ if the production functions of these tenants differ.¹³ This also implies that the rental percentage for different tenant farms may not be the same. That is, in maximizing the rental annuity of his landholdings, the owner will assign different land sizes and charge different rental percentages for different tenants if their production functions require different intensities of tenant inputs. In equilibrium, given homogeneous lands, the marginal productivity of land for each farm will be equal at every margin, for it is at the highest value of $(q - f)/h$ that land sizes are divided.

Third, the farming cost (other than the cost of land) may be shared by the tenant and the landowner jointly. In this case, the f/h curve represents the combined cost. Given q/h and f/h , as in Figure 4, the f/h minus the landowner's input cost will be lower, thus defining a higher $(q - f)^*/h$ curve. This higher $(q - f)^*/h$ curve (not drawn) measures not only the cost of land (rent) but also the landowner's non-land farming cost.¹⁴ The rental percentage charged by the landowner will accordingly be higher, with an upward shift of $(\partial q/\partial h)r$ by the same percentage at every point. The implication of this is important: it does not matter whether the landowner stipulates

¹³ The term "land homogeneity" employed here, which presupposes identical market location, renders the unambiguous condition: Given the same maximum values of $(q - f)/h$ and the same ratios of tenant inputs to land on different farms, the rental percentages will be the same. Different production functions associated with different crops will require different intensities of tenant inputs. On the other hand, given the production function, the rental percentage and the land size per tenant will depend on (1) the fertility of land and (2) the cost of tenant inputs.

¹⁴ Note that the peak of this higher $(q - f)^*/h$ curve will be to the left of T_1 . However, this is not relevant for land size division, for it will lead to a lower rent per acre. The dividing line is still T_1 , where $(q - f)/h$, or the net return to land, is at a maximum. But the rental percentage in this case will be $(q - f)^*/h$ at T_1 divided by q/h .

that the tenant is to invest more in land and charges a lower rental percentage or whether the landowner invests in land himself and charges the tenant a higher rental percentage; the investment will be made if it leads to a higher rental annuity.¹⁵ It follows that for any contract the tenant does not have to possess the required amount of input. If what he has is insufficient, the tenant may raise the farming inputs by co-operating with the landowner, by subleasing, by hiring farmhands, by borrowing, or by joint tenancy with another family. Moreover, the different tenant input requirements in farms of different land grades and production functions will match tenants with different input endowments accordingly.

We now review the solution conditions. The highest value of rent per unit of land, $(q - f)/h$, is derived when the marginal tenant cost equals the associated marginal product of tenant input. As illustrated in Figure 4, given the highest value of $(q - f)/h$, the land size per tenant (T_1) is determined, and the corresponding rental percentage, r , will be ar/ap . At the dividing line T_1 , the rent per acre of land, that is, ar , $(q - f)/h$ or rq/h , equals the marginal product of land, $\partial q/\partial h$. At T_1 , we also have the unique condition of the rental percentage

$$r = \frac{\partial q/\partial h}{q/h} = \frac{(q - f)/h}{q/h} = \frac{q - f}{q}.$$

This is identical with our mathematical solution, where we have

$$r = \frac{\partial q/\partial h}{q/h} = \frac{q - Wt}{q},$$

since in our mathematics we assume only one tenant input, $f = Wt$.

V. Comparison to Fixed Rent and the Market Criteria of Share Contracting

What are the differences between fixed and shared rent contracts? The basic difference between fixed rent and share tenancy lies in how the chosen labor-land ratio (or the ratio of non-land inputs to land) is expressed. With fixed rent, the tenant states how much land he will employ, and he alone decides—subject to the constraint of competition as with share tenancy—the amount of non-land inputs to be committed for every production run. Under share tenancy, the landowner and the tenant mutually decide the intensity of non-land inputs over land. In either case, the wealth-maximizing value determines the land size per farm as well as other inputs

¹⁵ The fact that most leases under share tenancy are not perpetual has been attacked as insecure tenure which leads to misallocation of resources. But it is no more insecure than hired farmhands who are subject to dismissal in case of poor performance. Indeed, a temporary lease is an effective device to insure efficient allocation. The argument that non-perpetual leases tend to discourage investment is also incorrect. An analysis on the choice of lease durations is offered in Cheung (in press, chap. iv, Section C).

employed. Since the set of constraints for decision making is the same for the two types of contracts, the same resource use is implied.

To illustrate, return momentarily to Figure 4. In a fixed rent contract, the landowner will be able to charge a rent equal to the maximum of $(q - f)/h$ per acre, or ar , and the area rented by the tenant will again be T_1 , where the rent per acre equals the marginal product of land. Given q/h and f/h as shown, if an area larger than T_1 is rented, the rental price per acre will be higher than the marginal product of land. Conversely, if an area smaller than T_1 is rented, the tenant's total income from farming will be less than his alternative earning. Thus, under fixed rent, the land size per farm and the rental annuity received by the landowner will be exactly the same as under share tenancy.¹⁶ Furthermore, if the same crops are to be planted on the same grade of land, having the same production function, the tenant under fixed rent will have to commit the same f/h to survive competition. As in sharecropping, if one tenant does not possess the necessary amount of inputs, he may acquire them through various means or rent a farm which requires less farming inputs.

Let us press further the question: Under share tenancy, what criteria will the landowner use in arriving at the equilibrium land size and rental percentage to each tenant? Clearly, the most efficient type of crops to be planted on the given land must be chosen, and the tenant inputs required must be considered—all *ex ante* decisions with some degree of uncertainty. Surely erroneous decisions may occur due to wrong expectations by either party of the contract. This will result in the tenant land size or the rental percentage differing from wealth maximization. However, it is equally clear that some simple rules exist in the market which the contracting parties will follow. Given land of a particular grade, there exists, at prevailing market prices of rent and other factors of production, certain combinations of tenant input requirements, rental percentages, and tenant land sizes.

Indeed, given freely alienable (marketable) land rights under private ownership, a landowner does not have to know the details of farming himself. Competing for resource ownerships will induce efficient contracts. If a less valuable crop is planted by the tenant, if the rental percentage is too low, the land size per tenant too large, or the tenant inputs required are too little, the rental annuity as a return to land for the landowner will be at a rate lower than the rate of interest. In this case, the landowner will either make the proper adjustments, lease the land to a different tenant, choose a different contractual arrangement, or sell his landownership outright.

¹⁶ Empirically, we find share rent being generally slightly higher than fixed rent (Cheung, in press, Appendix B). This slight difference might be explained as a payment to the landowner for his "risk" bearing under a share contract. The observed stipulations for fixed rent and share contracts also conform with our analysis (*ibid.*, chap. iv, Section B).

On the other hand, if the contractual agreements are such that the tenant is receiving a share lower than his alternative earning, other landowners will bid his services away.

As with competition among landowners, competition among tenants will insure that the contracted amount of tenant inputs be committed. Indeed, by checking the output alone the landowner will know whether the contracted terms are respected by the tenant and will decide whether the share contract should be continued. In practice, however, the transactions are frequently handled by hired agents who specialize in approximating the true yield. And in spite of numerous claims by reformers and novelists that tenants could make no decision and therefore were exploited, it is refreshing to point out that, in China at least, share contracting had been rather polite:

As soon as the grain is threshed the tenant invites the landlord to a feast, after which the grain is divided and the tenant delivers the landlord's share to his home. After it is properly delivered the landlord gives the tenant a feast. If the landlord wishes to dismiss the tenant he does so at this time. If the tenant wishes to leave, he takes the sails from the windmill just before the landlord comes to the tenant's feast as a sign to the landlord of his intentions [Buck, 1930, pp. 148–49].

VI. Some Concluding Remarks

Contrary to the prevailing view, standard economic theory shows that the use of privately owned resources under share tenancy satisfies the Pareto condition. Even if we relax the assumption of zero contracting cost, inefficient allocation does not necessarily follow (Cheung, in press, chap. iv). In the tax-equivalent analysis, the writers fail to realize that the percentage shares and area rented under share tenancy are not mysteriously “fixed” but are competitively determined in the market. Thus, such an analysis fails on several counts. First, in the traditional analysis, the possibility of the landlord's allocating his total holdings to *several* tenants has not been treated in an equilibrium framework. While this is valid in analyzing an excise tax, under share tenancy the cost and the distribution of land are thereby ignored. Second, the percentage share has been taken as given. Under share tenancy, however, the rental percentage is a discretionary variable. Third, in the case of a tax, the government is not contracting to maximize wealth.

We cannot analyze the way a person uses resources without first specifying the nature of his property rights. It is true that, once the land size and the rental percentage are contracted to a tenant, he prefers to work or invest less in land than if he cultivates his own land. But under private

ownership of land, the landlord's incentive to maximize his wealth is not reduced. We may well repeat a conclusion reached earlier: It does not matter whether the landowner stipulates that the tenant is to invest more in land and charges a lower rental percentage or whether the landowner invests in land himself and charges the tenant a higher rental percentage; the investment will be made if it leads to a higher rental annuity.

If economic efficiency is viewed as a condition of market equilibrium logically deduced from the theory of choice under private property rights, devoid of welfare implications, one wonders why the inefficiency argument of share contracts has prevailed for so long. Perhaps the illusion rendered by the *prima facie* similarity between an excise tax and a share contract offers a marginal equality which appears attractive. Or perhaps the frequent condemnation of sharecropping in land-tenure literature yields an impression of misallocation which appears convincing. I have tested implications of alternative hypotheses against observations chosen from periods and locations where private property rights prevail in agriculture (Cheung, in press, chaps. iii and iv). Not only do the observations confirm the theory derived here, but they refute the implications of the tax-equivalent analysis.

However, the inefficiency argument against share leases is only one of several which have been used against tenant farming in general. For example, a high rent is taken as "exploitation" which dampens the tenant's incentive, and short-term leases are regarded as "insecure" tenure which inevitably reduces investment in land. Unfortunately, these assertions have been frequently taken as facts. And measures of agrarian reform have been undertaken by governments to rule sharecropping illegal, to intervene in the terms of the contract, or to abolish farm tenancy altogether.¹⁷

In conjunction with the present study, a number of issues may be further explored. Not only are share contracts observed in agriculture, but they are also common among retail stores, beauty salons, gasoline stations, amusement park rentals, and even the much regulated oil and fishery industries. To the extent that investigation reveals that the constraints of competition differ, modifications need be made in the theory of share tenancy to interpret resource allocation in these industries. Furthermore, we may apply the theory to a situation where the ownership of some resource in a share contract belongs to the state. For example, in Taiwan before 1951, a portion of farmland owned by the government was leased to tenants on a share basis; in mainland China in the 1950's, a popular form of business firms involved state-private co-operation with sharing arrangements.

¹⁷ For a general discussion of land reforms in Asia, see Cheung (in press, chap. i). An analysis of the maximum rental share restriction enforced in Taiwan is seen in *ibid.*, chaps. v–viii.

Finally, an analysis of the choice of contracts is not offered here. Under private ownership of resources, the contracting parties are free to choose among a wage contract, a fixed rent contract, a share contract, or a combination of each. Why do share contracts usually predominate in agriculture? What determines the choice of contract? Why do the patterns of contractual arrangements differ greatly in various localities? These questions may be answered in one general analytical framework, and I have advanced elsewhere a choice-theoretic approach to explain contractual behaviour based on risk aversion subject to the constraint of transaction cost (Cheung, in press, chap. iv). Whatever the contractual arrangements, one of the main conclusions reached in this study is that these arrangements do not imply different efficiencies of resource allocation as long as property rights are exclusive and transferable.

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